Sheet 3

Effect probabilities and three paths

1. For $\rho = (\mathbb{1} + \sigma_x/3 + \sigma_z/2)/2$ and $F = (\mathbb{1} + (2/5)\sigma_x)/2$, check positivity of the state and effect, then compute $\mathrm{Tr}(\rho F)$ and the mean of $2F$. Is $2F$ itself an effect?
2. For the qubit projection assignment $h(\mathbf{n}) = (1 + n_z^5)/2$, verify the antipodal rule. Values on the coordinate axes would force which Bloch vector for a density representation? Test that representation at $n_z = 1/\sqrt{2}$. Explain which premise of Busch's theorem is absent.
3. In $\mathbb{C}^3$ prepare $(e_A + e_B + e_C)/\sqrt{3}$, keep evolution fixed at the identity and detect $(e_A + e^{i\phi} e_B + e_C)/\sqrt{3}$. Compute all seven nonempty-path probabilities per original trial, without renormalising filtered states. Verify $I_3 = 0$ for arbitrary ϕ, and list the probabilities at $\phi = \pi$.